

ECON0019: Quant Econ and Econometrics

Introduction to Time Series (Week 6)

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What to expect

In the last 5 weeks of the course, we'll cover a variety of topics, with a focus on time series data:

- This week: Introduction to time series
- Time series regression
- Forecasting and prediction
- Dynamic causal effects
- Critical assessment of econometric studies

Logistics: feedback hours Tuesdays 3-4pm in Drayton House 202
Empirical Project now available on Moodle – discussed in tutorials.

An unbiased case for (time series) econometrics

The material in this module is some of the most useful you will learn in your Economics course.

Data analysis skills have never been in higher demand - finance, consulting, tech, public sector, academia.

At the very least, need no be a critical consumer of data - abused more than ever!

Time series data - and forecasting/prediction - is one of the most sought after skills within econometrics.

The final quarter of this module aims to give you practical - and marketable - skills.

Time Series data

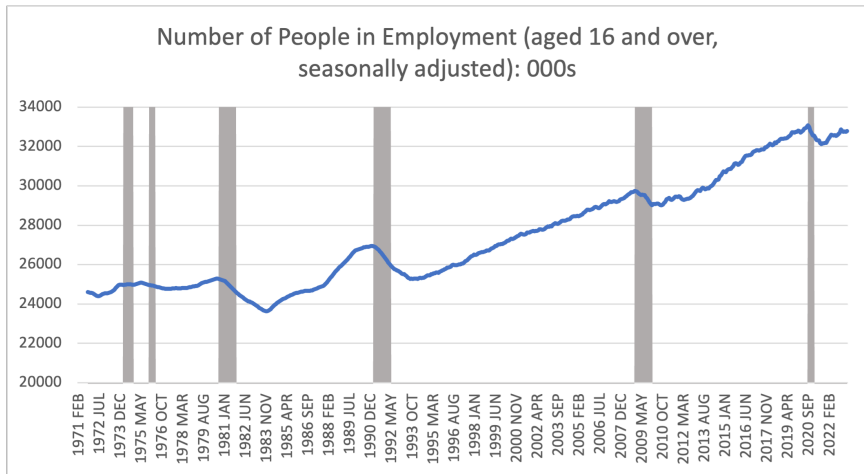
Previously in the course you have studied data featuring observations of many units – time series is different!

- In panel data, you considered many units with repeated observations.
- In time series data, a single unit is observed repeatedly (quarterly GDP, daily exchange rate,...).
- This poses important new challenges for causal and statistical inference.

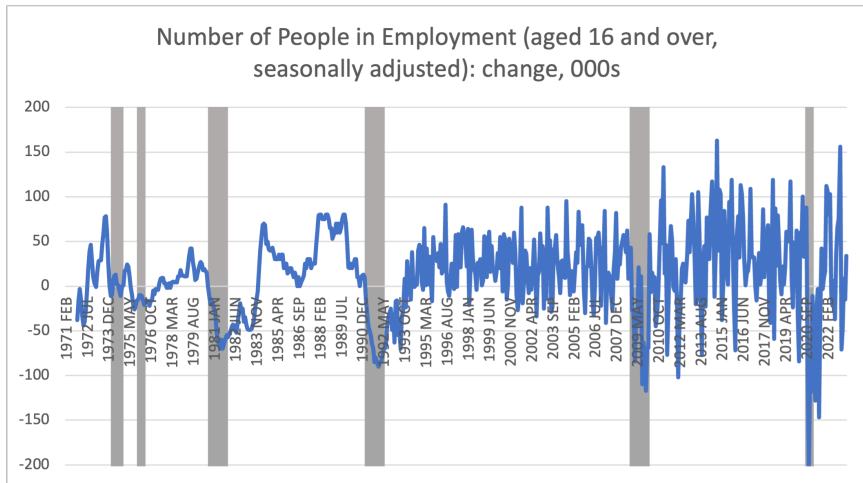
So far you have focused on causal analysis – now, we'll also discuss forecasting and prediction.

Today: time series data. Next: forecasting/prediction, and then dynamic causal effects.

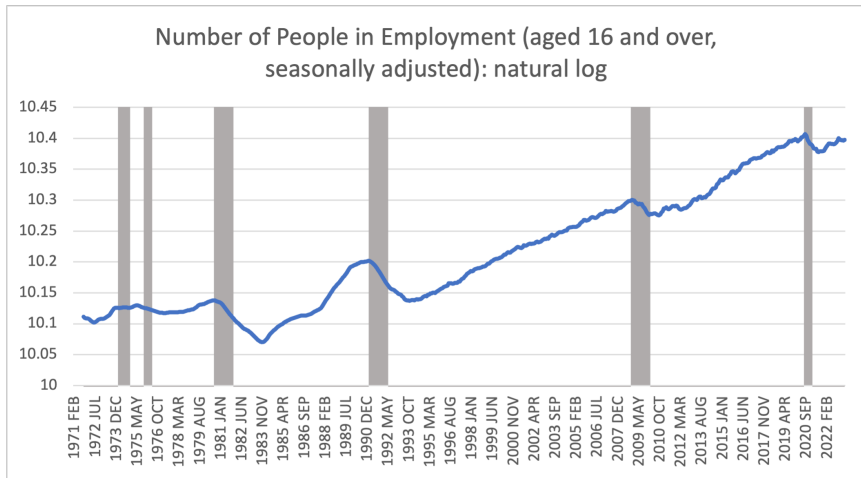
Some common time series



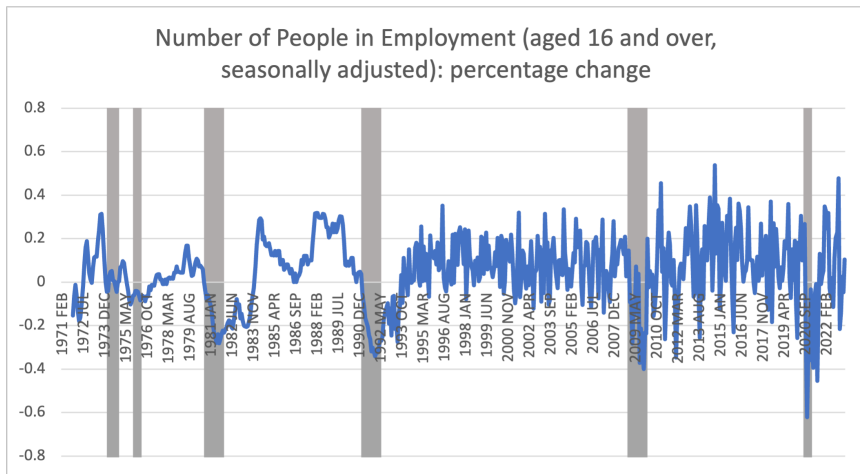
Some common time series



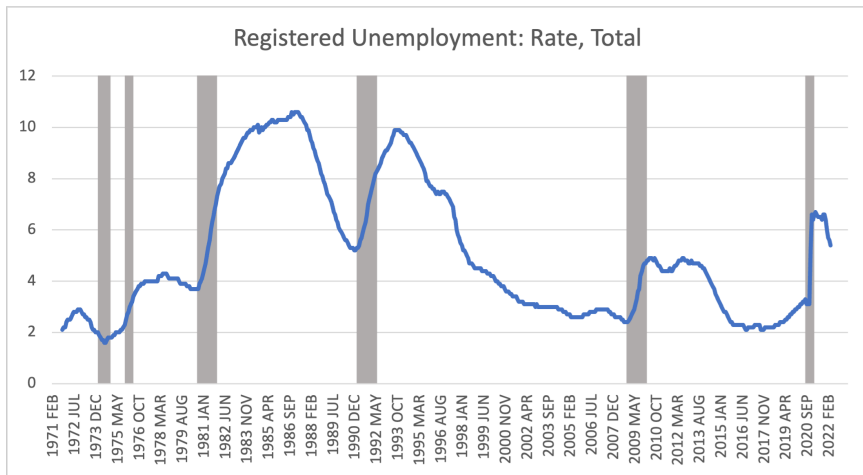
Some common time series



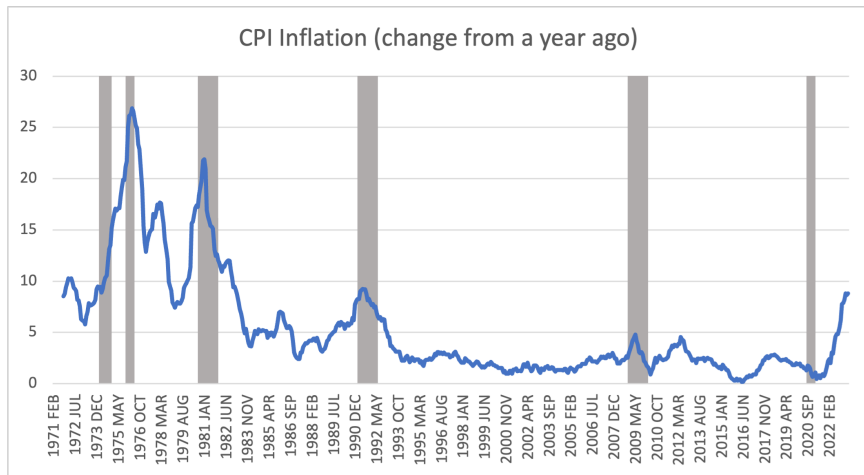
Some common time series



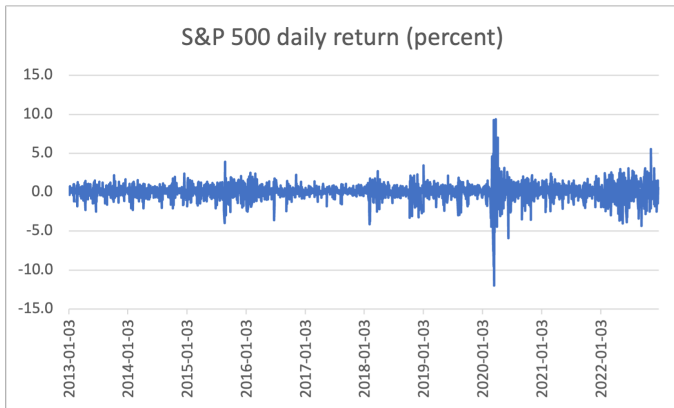
Some common time series



Some common time series



Some common time series



Why time series?

- Forecasting
- *Dynamic* causal effects
- Modeling risks
- More broadly: environmental and climate modeling, engineering (system dynamics), computer science (network dynamics)

Technical challenges

Time series data poses new important technical challenges:

- We don't get to observe a random sample of individuals, just repeated observations of one individual. Under what conditions can we learn from them?
- Observations are *correlated* (what about LS assumptions?).
- Time lagged variables are relevant regressors.
- Errors are serially correlated (how to compute standard errors?).
- Samples can be short (annual or quarterly data).

Some notation

- y_t is the value of y in period t .
- Data set: $\{y_1, \dots, y_T\}$ are T observations of the time series of y .
- The **first lag** of a time series y_t is y_{t-1} ; **j^{th} lag** is y_{t-j} .
- The **first difference** of a series, Δy_t , is its change between periods $t-1$ and t , that is, $\Delta y_t = y_t - y_{t-1}$.

Some notation

- The first difference of the logarithm of y_t is $\Delta \ln(y_t) = \ln(y_t) - \ln(y_{t-1})$.
- The percentage change of a time series between periods $t-1$ and t is approximately $100\Delta \ln(y_t)$.
- Important distinction: percent/percentage change vs. percentage point change.
- For our purposes, we only consider consecutive, evenly-spaced observations.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Example: quarterly rate of GDP growth at an annual rate

GDP = Real GDP in the US (Billions of \$2009)

- GDP in the fourth quarter of 2016 (2016:Q4) = 16851
- GDP in the first quarter of 2017 (2017:Q1) = 16903
- Percentage change in GDP, 2016:Q4 to 2017:Q1

$$= 100 \times \left(\frac{16903 - 16851}{16851} \right) = 0.31\%$$

- Percentage change in GDP, 2012:Q1 to 2012:Q2, at an annual rate = $100 \times ((1 + 0.31/100)^4 - 1) = 1.25\%$
- $4 \times 0.31\% = 1.23\% \approx 1.25\%$
- Using the log approximation:

$$4 \times 100 \times (\ln(16903) - \ln(16851)) = 1.23\%$$

Example continued...

TABLE 15.1 GDP in the United States in the Last Quarter of 2016 and in 2017

Quarter	U.S. GDP (billions of \$2009), GDP_t	Logarithm of GDP, $\ln(GDP_t)$	Growth Rate of GDP at an Annual Rate, $GDPGR_t = 400 \times \Delta \ln(GDP_t)$	First Lag, $GDPGR_{t-1}$
2016:Q4	16,851	9.732	1.74	2.74
2017:Q1	16,903	9.735	1.23	1.74
2017:Q2	17,031	9.743	3.01	1.23
2017:Q3	17,164	9.751	3.11	3.01
2017:Q4	17,272	9.757	2.50	3.11

Note: The quarterly rate of GDP growth is the first difference of the logarithm. This is converted into percentages at an annual rate by multiplying by 400. The first lag is its value in the previous quarter. All entries are rounded to the nearest decimal.

Autocovariance and autocorrelation

The correlation of a series with its own lagged values is called **autocorrelation** or **serial correlation**.

- The first autocovariance of y_t is $cov(y_t, y_{t-1})$.
- The first autocorrelation of y_t is $corr(y_t, y_{t-1})$.

$$corr(y_t, y_{t-1}) = \frac{cov(y_t, y_{t-1})}{\sqrt{(var(y_t)var(y_{t-1}))}} \equiv \rho_1$$

- Similarly for the j^{th} autocovariance/correlation:

$$cov(y_t, y_{t-j}), \quad corr(y_t, y_{t-j})$$

Sample autocovariance

It's easy to estimate the autocovariance/correlation:

$$\hat{\rho}_j = \frac{\widehat{cov}(y_t, y_{t-j})}{\widehat{var}(y_t)},$$

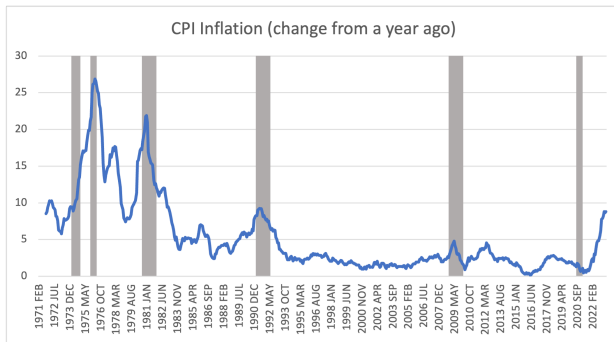
where

$$\widehat{cov}(y_t, y_{t-j}) = \frac{1}{T} \sum_{t=j+1}^T (y_t - \bar{y}_{j+1,T})(y_{t-j} - \bar{y}_{1,T-j})$$

and $\bar{y}_{t+j,T}$ is the sample average of y_t computed over observations $t = j+1, \dots, T$.

- Note: variance in denominator, summation bounds, and divisor

Autocorrelation of inflation



Lag Inflation rate ($\ln f_t$)

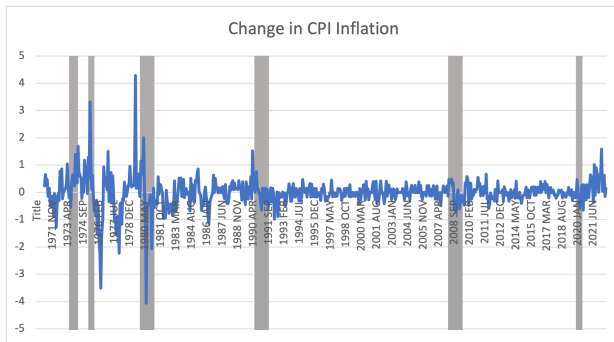
1 0.99

2 0.99

3 0.97

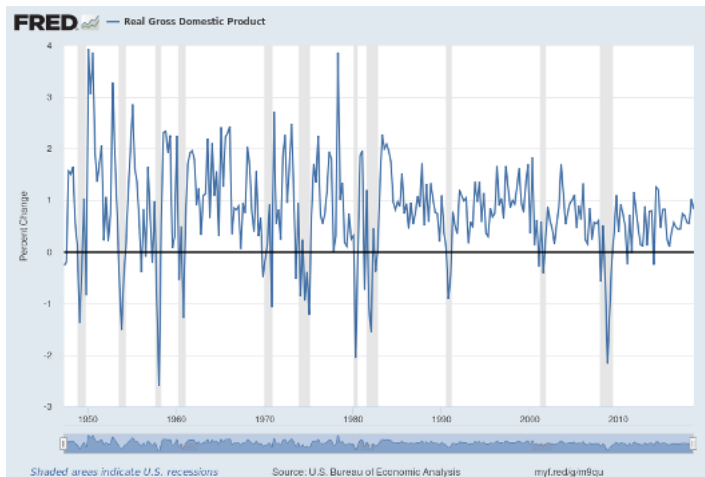
4 0.96

Autocorrelation of inflation



Lag	Inflation rate (Inf_t)	Change in Inflation rate (ΔInf_t)
1	0.99	0.42
2	0.99	0.35
3	0.97	0.23
4	0.96	0.14

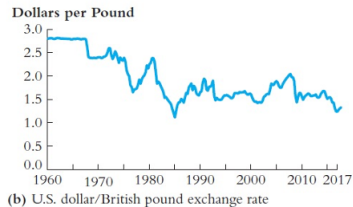
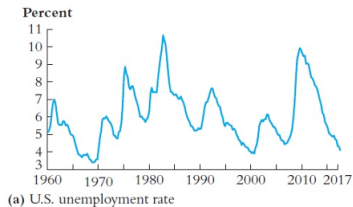
Some examples



$$\hat{\rho}_1 = 0.33, \hat{\rho}_2 = 0.26, \hat{\rho}_3 = 0.10, \hat{\rho}_4 = 0.11$$

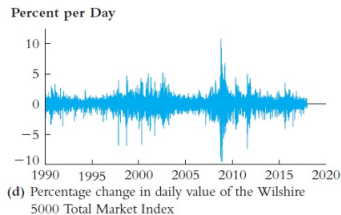
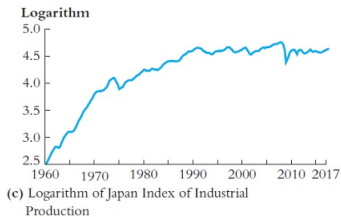
Some examples

Do these look autocorrelated?



Some examples

Do these look autocorrelated?



Review: the Least Squares assumptions

Remember from term 1:

- 1 MLR.1 Linear in parameters:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$$

- 2 MLR.2 (Random Sampling, i.i.d. data): We have a random sample of size n from the population,

$$(x_{1i}, x_{2i}, \dots, x_{ki}, y_i) : i = 1, \dots, n$$

- 3 MLR.3 (No Perfect Collinearity): In the sample, none of the explanatory variables is constant, and there are no exact linear relationships among them.
- 4 MLR.4 (Zero Conditional Mean, exogeneity):

$$E[u_j | x_1, x_2, \dots, x_k] = 0$$

for all $(x_1, \dots, x_k)$.

The Least Squares assumptions for time series data

Denote $X_t = (x_{1t}, x_{2t}, \dots, x_{kt})'$.

$$y_t = \beta_0 + X_t' \beta_1 + X_{t-1}' \beta_2 + \dots + X_{t-p}' \beta_{p+1} + u_t$$

Recall the Least Squares assumptions:

- MLR.1: Linear in parameters ✓
- MLR.3: No perfect collinearity ✓
- MLR.4: Zero conditional mean/exogeneity,
 $E[u_t | X_t, \dots, X_{t-p}] = 0$

We may strengthen exogeneity to **strict exogeneity**:

$$E[u_t | \dots, X_{t+1}, X_t, X_{t-1}, \dots] = 0$$

Stationarity

Recall MLR.2, the **i.i.d.** assumption - this was trivially satisfied with random sampling from a population.

Let's focus on the **identical** part - does this make sense in time series data?

Stationarity

Recall MLR.2, the **i.i.d.** assumption - this was trivially satisfied with random sampling from a population.

Let's focus on the **identical** part - does this make sense in time series data?

Time series counterpart is **stationarity**.

A time series y_t is stationary if its probability distribution does not change over time.

- In particular, the joint distribution of $(y_{s+1}, y_{s+2}, \dots, y_{s+T})$ does not depend on s . For example,

$$\begin{aligned}E[y_t] &= E[y_{t+1}] = E[y_{t+2}] \dots, \\E[y_t^2] &= E[y_{t+1}^2] = E[y_{t+2}^2] \dots, \\&\dots\end{aligned}$$

- Otherwise, y_t is **nonstationary**. What about existence?

Stationarity

- **Joint stationarity:** distribution of $(y_{s+1}, X_{s+1}, y_{s+2}, X_{s+2}, \dots, y_{s+T}, X_{s+T})$ does not depend on s .
- Stationarity is important for both practical and statistical reasons!
- Often, we will use “second order stationarity” meaning that all first and second moments are both time invariant and finite:

$$\begin{aligned} E[y_t] &= E[y_s], E[X_t] = E[X_s], E[y_t^2] = E[y_s^2], E[X_t X_t'] = E[X_s X_s'], \\ E[y_t y_{t-j}] &= E[y_s y_{s-j}], E[X_t X_{t-j}'] = E[X_s X_{s-j}'], \\ E[y_t X_{t-j}] &= E[y_s X_{s-j}], \forall t, s, j \end{aligned}$$

Stationarity is a natural counterpart for the “identically distributed” assumption from the familiar **Least Squares Assumptions** (MLR.2: Random sampling).

Independence?

But what about “independently distributed”?

Independence assumption is no longer plausible - why?

Independence?

But what about “independently distributed”?

Independence assumption is no longer plausible - why?

We observe all data (over a period) for a **single unit**.

Recall that data yesterday often *very* relevant to data today:
observations are not independent!

Instead, we'll rely on the idea that data *sufficiently far apart* can
be treated as if it is independent!

Weak dependence

Cannot assume independence of $(y_t, X_t, X_{t-1}, \dots)$ and $(y_{t-1}, X_{t-1}, X_{t-2}, \dots)$ – instead assume that dependence is **weak**.

Dependence dies out fast enough that two observations chosen adequately far apart in time are essentially independent.

One formalization is **ergodicity**:

$$\begin{aligned} & \lim_{n \rightarrow \infty} |E[g(y_t, \dots, y_{t+k})h(y_{t+n}, \dots, y_{t+k+n})]| \\ &= |E[g(y_t, \dots, y_{t+k})]| |E[h(y_{t+n}, \dots, y_{t+k+n})]| \end{aligned}$$

Must hold for (y_t, X_t) *jointly*.

Properties of OLS in time series

We end up with the five assumptions of Time Series regression:

- 1 TS1 Linear in parameters
- 2 TS2 Strict exogeneity / TS2' exogeneity
- 3 TS3 No perfect collinearity
- 4 TS4 Stationarity (e.g., covariance stationarity)
- 5 TS5 Weak dependence/ergodicity

Under TS1, TS2, TS3, OLS is **unbiased**.

Under TS1, TS2', TS3, TS4, and TS5, OLS is **consistent** and **asymptotically normal**.

What is the asymptotic variance?

(Strict) Exogeneity

OLS is not unbiased without **strict exogeneity**:

$$E[u_t | \dots, X_{t+1}, X_t, X_{t-1}, \dots] = 0.$$

We can no longer view the RHS variables as **predetermined**. We need u_t to be mean-independent of independent variables at all times. Why?

$$\hat{\beta} = \beta + \left(\sum_{t=p+1}^T \tilde{X}_t \tilde{X}_t' \right)^{-1} \sum_{t=p+1}^T \tilde{X}_t u_t,$$

where $\tilde{X}_t$ stacks $X_t, \dots, X_{t-p}$ and a constant.

Suppose $X_t = f(u_{t-1}, u_{t-2}, \dots) + e_t$, for example,

- $X_t = y_{t-1}$ (autoregression, next week)
- $X_t = \gamma_1 y_{t-1} + \gamma_2 y_{t-2} + \dots + e_t$ (reverse causality?)

In each case, X_{t+1} is a function of u_t !

What goes wrong

Only assuming exogeneity (TS2'),

$$\begin{aligned} E[\hat{\beta} - \beta] &= E \left[\left(\sum_{t=p+1}^T \tilde{X}_t \tilde{X}_t' \right)^{-1} \sum_{t=p+1}^T \tilde{X}_t u_t \right] \\ &= E \left[\left(\sum_{t=p+1}^T (f(u_{t-1}) + e_t)(f(u_{t-1}) + e_t)' \right)^{-1} \sum_{t=p+1}^T (f(u_{t-1}) + e_t) u_t \right] \\ &\neq \left(E \left[\sum_{t=p+1}^T (f(u_{t-1}) + e_t)(f(u_{t-1}) + e_t)' \right] \right)^{-1} E \left[\sum_{t=p+1}^T \tilde{X}_t u_t \right] \neq 0. \end{aligned}$$

Time series dependence causes correlation between the numerator and denominator!

We need **strict exogeneity** to guarantee that u_t is mean independent of X_t in the denominator for all past and future periods.

It says that all $\dots X_{t+1}, X_t, X_{t-1}, \dots$ cannot be a function of u_t .

Consistency

The argument for consistency is very similar to that under MLR, but using different results.

The **Ergodic Theorem** states that if Z_t is ergodic and stationary and $E[Z_t] = \mu$, then $\frac{1}{T} \sum_{t=1}^T Z_t - \mu \xrightarrow{P} 0$.

Under TS1, TS2', TS3-TS5, we can apply the Ergodic Theorem to show that

$$\frac{1}{T} \sum_{t=1}^T \tilde{X}_t y_t =$$

$$\frac{1}{T} \sum_{t=1}^T \tilde{X}_t \tilde{X}_t' =$$

$$\text{Finally, } \left(\frac{1}{T} \sum_{t=1}^T \tilde{X}_t \tilde{X}_t' \right)^{-1} \frac{1}{T} \sum_{t=1}^T \tilde{X}_t y_t$$

Asymptotic normality

Arguments for asymptotic normality also similar.

A central limit theorem for sequences of ergodic and stationary random variables shows that if Z_t is ergodic and stationary, then

$\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t \xrightarrow{d} \mathcal{N}(0, V)$, where

$$V = \lim_{T \rightarrow \infty} \frac{1}{T} E \left[\left(\sum_{t=1}^T Z_t \right) \left(\sum_{t=1}^T Z_t \right)' \right].$$

Then as before, $\hat{\beta} = \beta + \left(\frac{1}{T} \sum_{t=1}^T \tilde{X}_t \tilde{X}_t' \right)^{-1} \frac{1}{T} \sum_{t=1}^T \tilde{X}_t u_t$,

Is OLS BLUE?

Recall that under the additional assumption of homoskedasticity (Gauss-Markov assumptions) OLS was the BLUE.

In time series settings, homoskedasticity is not enough – also need error term to be **serially** uncorrelated.

If error term is serially correlated, can construct more efficient estimator.